

KAHULUI AP, HI, USA

WMO: 911900

Lat: 20.900N

Lon: 156.429W

Elev: 16

StdP: 101.14

Time Zone: -10.00 (W10)

Period: 94-19

WBAN: 22516

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.1	16.0	11.2	8.3	19.6	12.3	8.9	19.8	13.1	25.3	12.0	24.8	2.0	150	0.480

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.3	32.4	23.9	31.6	23.6	31.0	23.3	25.3	30.0	24.7	29.5	24.3	29.1	7.8	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	18.9	27.2	23.4	18.2	26.9	22.8	17.6	26.7	77.2	30.0	74.9	29.6	73.0	29.1	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
12.0	11.3	10.7	DB	12.4	33.7	1.5	1.2	11.3	34.5	10.5	35.2	9.6	35.9	8.6	36.7
			WB	11.2	25.9	1.2	1.3	10.3	26.8	9.6	27.6	8.9	28.4	8.0	29.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.5	22.3	22.3	22.8	23.7	24.5	25.6	26.3	26.6	26.4	25.7	24.4	23.1
	DBStd	2.00	1.43	1.40	1.36	1.35	1.20	1.00	1.08	1.02	1.23	1.19	1.31	1.31
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	5287	381	344	396	410	448	469	506	515	492	488	432	406
	CDD18.3	2245	122	111	137	160	190	219	247	257	242	230	182	148
	CDH23.3	16476	631	556	714	959	1372	1746	2139	2303	2204	1902	1189	761
CDH26.7	4569	66	61	98	192	360	540	711	792	775	608	269	97	
Wind	WSAvg	5.5	4.6	4.8	5.1	5.7	5.5	6.5	6.6	6.4	5.7	5.3	5.1	4.8
Precipitation	PrecAvg	431	87	64	67	33	19	4	12	13	10	26	58	74
	PrecMax	882	367	211	277	113	77	20	42	60	36	144	236	259
	PrecMin	135	0	2	3	0	0	0	0	0	0	0	2	0
	PrecStd	191	85	55	58	30	20	4	10	12	10	29	54	61
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.4	29.3	29.6	30.6	31.8	32.3	33.0	33.2	32.7	31.3	29.9	
		MCWB	21.7	21.9	21.8	22.4	23.0	23.2	24.2	24.6	24.2	24.0	23.5	22.5
	2%	DB	28.2	28.2	28.6	29.5	30.4	31.0	31.8	31.9	32.1	31.6	30.1	28.6
		MCWB	21.3	21.5	21.6	22.0	22.4	22.8	23.8	23.8	23.9	23.6	23.0	21.8
	5%	DB	27.4	27.4	27.8	28.6	29.5	30.2	30.8	31.1	31.2	30.7	29.2	27.8
		MCWB	21.1	21.1	21.3	21.7	22.0	22.5	23.3	23.5	23.6	23.4	22.7	21.5
	10%	DB	26.6	26.6	27.0	27.8	28.7	29.5	30.1	30.4	30.4	29.9	28.3	27.0
		MCWB	20.8	20.7	20.9	21.4	21.9	22.3	22.9	23.2	23.2	23.0	22.3	21.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.5	23.6	23.9	24.6	24.4	24.5	25.9	26.1	26.1	25.4	24.9	24.3
		MCDB	26.6	27.2	27.3	27.7	28.9	30.0	30.2	30.9	30.9	30.2	29.0	27.6
	2%	WB	22.8	22.7	22.9	23.1	23.7	23.9	24.9	25.1	25.1	24.7	24.1	23.2
		MCDB	26.1	26.3	26.4	27.3	28.5	29.3	29.6	30.0	29.9	29.5	28.2	26.6
	5%	WB	22.1	22.1	22.2	22.4	23.0	23.4	24.2	24.5	24.5	24.1	23.5	22.5
		MCDB	25.7	25.7	26.0	26.9	27.7	28.7	29.0	29.3	29.5	28.8	27.6	26.0
	10%	WB	21.5	21.5	21.6	21.9	22.5	22.9	23.7	24.0	23.9	23.6	22.9	21.9
		MCDB	25.2	25.2	25.4	26.5	27.2	28.2	28.5	28.7	29.0	28.4	26.9	25.4
Mean Daily Temperature Range	5% DB	MDBR	8.9	8.7	8.5	8.4	9.0	8.5	8.2	8.3	8.9	8.9	8.3	8.3
		MCDBR	10.2	10.3	10.0	9.9	10.3	9.4	9.2	9.0	9.7	9.9	9.7	9.7
		MCWBR	5.3	5.3	5.0	4.6	4.6	4.1	3.9	4.0	4.2	4.5	4.7	5.0
	5% WB	MCDBR	8.8	8.7	8.9	9.1	9.3	8.8	8.5	8.6	9.1	9.4	8.8	8.2
		MCWBR	4.9	4.8	4.9	4.6	4.6	4.1	4.0	4.1	4.3	4.7	4.7	4.8
Clear-Sky Solar Irradiance	taub	0.390	0.398	0.410	0.423	0.413	0.398	0.399	0.399	0.389	0.400	0.401	0.382	
	taud	2.470	2.445	2.419	2.364	2.429	2.471	2.478	2.478	2.505	2.487	2.473	2.519	
	Ebn at Noon	882	896	896	884	883	889	889	894	903	883	864	879	
	Edh at Noon	103	111	118	127	118	112	111	111	107	105	102	95	
All-Sky Solar Radiation	RadAvg	4.48	5.05	5.66	6.46	6.88	7.08	7.00	6.80	6.36	5.39	4.55	4.10	
	RadStd	0.19	0.30	0.53	0.38	0.35	0.18	0.17	0.28	0.18	0.14	0.11	0.35	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (12 neighbors)	N/A	N/A	N/A	+0.55	N/A	N/A	N/A	N/A	+195	+194
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+90	+90

Nomenclature: See separate page